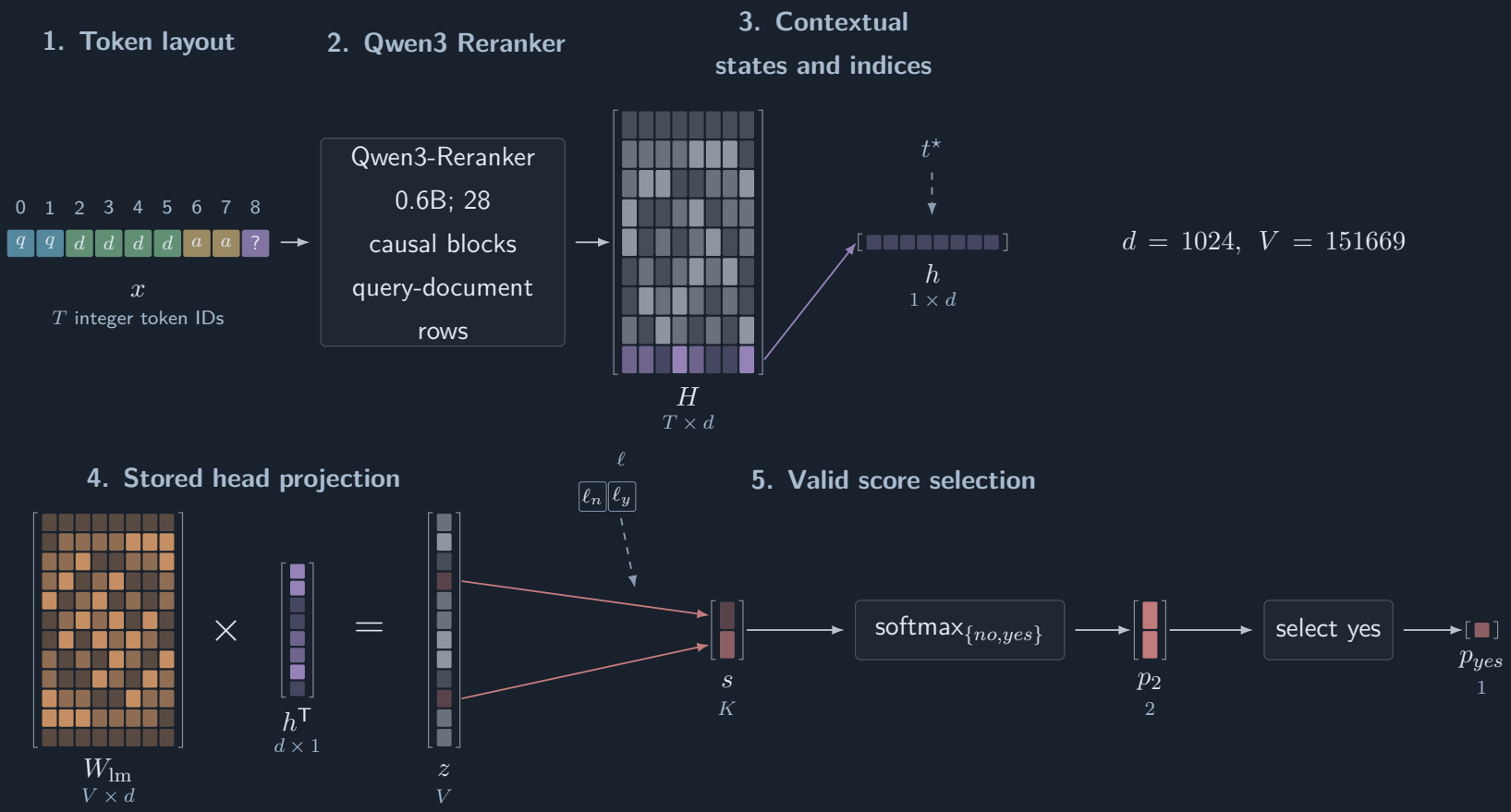
$$z = W_{\text{lm}} h^{\mathsf{T}}, \quad p_{yes} = \text{softmax}(z_{no}, z_{yes})_{yes} = \sigma(z_{yes} - z_{no})$$



Axes	$d = 1024$, $V = 151669$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.
Objects	Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.
Mechanism	P(yes) is a pointwise relevance score; no softmax over documents. Q has width 16x128=2048, K/V width 8x128=1024; hidden width is not the head concatenation width. In the index cells, n=no and y=yes. Official Transformers inference: full vocabulary logits, gather no/yes, normalize those two values. This does not identify the exact training token mask or EOS objective.